

Chapter 3 - Research Methodology

Preliminary Guide-Review Version

Status: This is an interim methodology and data record based on the first 24 Google Forms submissions. The approved Chapter 3 in the main Version 3 folder remains the thesis methodology.

3.1 Research Design

The study uses a quantitative, descriptive, correlational, and cross-sectional survey design. Employees answer with reference to one programme attended during the previous 12 months that combined digital learning with face-to-face learning. The unit of analysis is the individual employee.

Hybrid Learning -->

++> Learner Engagement --> Training Effectiveness

Trainer Competence -->

The arrows show relationships to be examined in employee responses. They do not establish a causal process or formal mediation effect.

3.2 Objectives and Hypotheses

The study describes employees' perceptions of Hybrid Learning, Trainer Competence, Learner Engagement, and Training Effectiveness. It examines the relationship of Hybrid Learning and Trainer Competence with Learner Engagement and the relationship between Learner Engagement and Training Effectiveness.

Hypothesis	Relationship examined
H1	Hybrid Learning has a positive relationship with Learner Engagement.
H2	Trainer Competence has a positive relationship with Learner Engagement.
H3	Learner Engagement has a positive relationship with Training Effectiveness.

3.3 Population, Interim Sample and Eligibility

The population is employees connected with Union Bank of India Regional Office Chennai South who attended a programme containing both digital or online learning and face-to-face learning during the previous 12 months. Purposive sampling is used because employees who experienced only one delivery mode cannot assess the connection between the two modes.

The first Google Forms export contained 24 records. The screening question asked whether the respondent attended a programme using both modes. Twenty-two respondents selected Yes and two selected No. The No records remain in the raw-data sheet for transparency but are excluded from every score and statistical test.

Screening stage	Records
Submitted Google Forms records	24
Excluded: selected No on eligibility question	2
Eligible records retained for preliminary analysis	22

This is an interim sample, not the planned final sample of approximately 80 eligible employees.

3.4 Questionnaire and Coding

The questionnaire contains four eligibility or profile questions and 26 five-point agreement items. Strongly Disagree is coded 1, Disagree is coded 2, Neither Agree nor Disagree is coded 3, Agree is coded 4, and Strongly Agree is coded 5. Higher scores show a more positive response.

Construct	Item numbers	Number of items	Score
Hybrid Learning	Q5-Q11	7	Arithmetic mean of Q5-Q11
Trainer Competence	Q12-Q17	6	Arithmetic mean of Q12-Q17
Learner Engagement	Q18-Q23	6	Arithmetic mean of Q18-Q23
Training Effectiveness	Q24-Q30	7	Arithmetic mean of Q24-Q30

Union Vidya is the digital-platform context where it was used in a selected programme. It is not a separate scored construct.

3.5 Data Quality and Analysis

The Google Form does not request names, employee numbers, customer information, account details, or other confidential Bank information. Results are reported only in aggregate form.

The preliminary check found no missing Likert values among the 22 eligible records, no invalid response codes, and no exact duplicate records. Five eligible records used one response category for all 26 items. Four selected Strongly Agree for all items and one selected Strongly Disagree for all items. The records are retained because the available data do not prove that they are invalid. They are recorded as a limitation.

Frequencies and percentages describe the profile. Construct means and standard deviations describe responses. Cronbach's alpha assesses internal consistency. Pearson correlation assesses pairwise association. Two preliminary regressions are calculated:

$$\text{Learner Engagement} = b_0 + b_1(\text{Hybrid Learning}) + b_2(\text{Trainer Competence}) + e$$

$$\text{Training Effectiveness} = b_0 + b_1(\text{Learner Engagement}) + e$$

3.6 Interpretation Boundary

The 22 eligible records cannot provide final dissertation findings. The sample is small, non-probability based, and self-reported at one point in time. The analysis cannot establish causation, job performance, organisational results, or formal mediation. Final analysis will repeat the same process after the intended eligible sample is collected.